

Caputo Dirichlet Heat Flow: Sharp Smoothing, Schatten, and Resolvent Limits

Route-A source-local certificate HCS-C277

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Abstract

We solve the Caputo time-fractional heat equation on $(0, \pi)$ in its exact Dirichlet sine basis and prove inverse-stable subordination, positivity, and contraction for every $0 < \beta \leq 1$. No source operator ideal is promoted to target arithmetic data.

1 The frozen memory owner

Let $X = L^2(0, \pi)$ and

$$A = -\frac{d^2}{dx^2}, \quad D(A) = H^2(0, \pi) \cap H_0^1(0, \pi).$$

The normalized eigenvectors $e_n = (2/\pi)^{1/2} \sin(nx)$ satisfy $Ae_n = n^2 e_n$. We freeze

$${}^C D_t^\beta u(t) + Au(t) = 0, \quad u(0) = u_0 \in X, \quad 0 < \beta \leq 1. \quad (1)$$

For $0 < \beta < 1$,

$${}^C D_t^\beta f(t) = \frac{1}{\Gamma(1-\beta)} \int_0^t (t-s)^{-\beta} f'(s) ds,$$

and $\beta = 1$ means the ordinary derivative. Bounded-domain fractional diffusion-wave problems admit eigenfunction constructions of this type [2]; here the interval permits sharp operator thresholds.

Writing

$$E_\beta(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\beta k + 1)},$$

the coefficient identity

$${}^C D_t^\beta \frac{t^{\beta k}}{\Gamma(\beta k + 1)} = \frac{t^{\beta(k-1)}}{\Gamma(\beta(k-1) + 1)}, \quad k \geq 1,$$

gives the unique solution family

$$S_\beta(t)u_0 = \sum_{n \geq 1} E_\beta(-n^2 t^\beta) \langle u_0, e_n \rangle e_n. \quad (2)$$

Pollard proved that $E_\beta(-x)$ is completely monotone on $x \geq 0$ for $0 < \beta < 1$ [1]. More precisely, if $\eta_\beta(t, s)$ is the inverse-stable density characterized by

$$\int_0^\infty e^{-zt} \eta_\beta(t, s) dt = z^{\beta-1} e^{-sz^\beta},$$

then scalar Laplace inversion in (2) yields

$$S_\beta(t) = \int_0^\infty e^{-sA} \eta_\beta(t, s) ds. \quad (3)$$

Thus $S_\beta(t)$ is self-adjoint and positive, preserves nonnegative functions, and is contractive. Dominated convergence in (2) gives strong continuity at zero.

References

- [1] H. Pollard, *The completely monotonic character of the Mittag-Leffler function $E_a(-x)$* , Bull. Amer. Math. Soc. 54 (1948), 1115–1116, [doi:10.1090/S0002-9904-1948-09132-7](https://doi.org/10.1090/S0002-9904-1948-09132-7).
- [2] K. Sakamoto and M. Yamamoto, *Initial value/boundary value problems for fractional diffusion-wave equations and applications to some inverse problems*, J. Math. Anal. Appl. 382 (2011), 426–447, [doi:10.1016/j.jmaa.2011.04.058](https://doi.org/10.1016/j.jmaa.2011.04.058).